

FROM GENE MUTATION TO DISEASE **(Duchene muscular dystrophy)**

Background

Duchenne muscular dystrophy (DMD) is an inherited muscular disorder caused by pathogenic variants in the DMD gene. It is characterized by progressive muscle weakness and primarily affects skeletal and cardiac muscles. DMD is inherited in an X-linked recessive pattern and occurs much more frequently in males.

The DMD gene provides instructions for producing dystrophin, a large protein that is important for maintaining the stability and protection of muscle fibers. Dystrophin is part of a protein complex that connects the internal structural framework of muscle cells with proteins associated with the extracellular environment. This connection helps protect muscle fibers from damage during repeated contraction and relaxation. Mutations that greatly reduce or prevent the production of functional dystrophin can result in Duchenne muscular dystrophy.

In this study, the DMD gene was investigated through a bioinformatics workflow using a reference DMD transcript obtained from the National Center for Biotechnology Information (NCBI). An artificial mutation was introduced into the sequence to examine how a premature stop signal could affect the predicted dystrophin protein. The analysis focused on comparing the original protein sequence with the artificially modified protein sequence.

Gene and Normal Protein Function

The gene investigated in this study was the DMD gene, which encodes the dystrophin protein. The DMD gene is located on the X chromosome and is one of the largest known human genes. Dystrophin is expressed mainly in skeletal and cardiac muscle, where it contributes to the structural stability of muscle fibers.

The reference sequence used in this study was:

NCBI accession: NM_004006.3

Organism: Homo sapiens

Gene: DMD (dystrophin)

Transcript: DMD transcript variant Dp427m, mRNA

The sequence was obtained from the NCBI reference record and processed in Galaxy. The coding sequence was translated to obtain the corresponding protein sequence. This protein sequence served as the original or wild-type (WT) sequence for comparison with the artificially modified sequence.

Dystrophin is a very large protein with multiple functional regions. Its structure allows it to participate in a protein complex that supports the mechanical stability of muscle cells. Loss of functional dystrophin can make muscle fibers more vulnerable to damage.

Documented Mutations Associated with DMD

Duchenne muscular dystrophy can result from many different pathogenic variants in the DMD gene. More than 2,000 mutations have been identified in association with Duchenne and Becker muscular dystrophy. These include deletions, duplications, and smaller changes in the DNA sequence.

Large deletions are among the most common types of DMD mutations. Other variants can alter individual nucleotides and may change the resulting protein. Some mutations disrupt the reading frame or introduce premature termination signals, which can prevent production of a functional dystrophin protein.

The mutation created in this project is not a documented patient mutation. It was deliberately introduced as an artificial mutation for the purpose of demonstrating the effect of a premature stop signal. Therefore, the artificial mutation should not be interpreted as a naturally occurring clinical DMD variant.

Hypothesis

If an artificial mutation introduces a premature stop codon into the DMD coding sequence, then translation of the modified sequence will terminate earlier than in the original sequence, producing a truncated dystrophin protein that differs from the normal protein.

Methods

- Retrieval of the DMD Sequence

The DMD reference transcript was obtained from NCBI using accession number NM_004006.3, corresponding to *Homo sapiens* dystrophin (DMD), transcript variant Dp427m, mRNA. The coding sequence was selected for subsequent sequence analysis.

- Sequence Processing

The DMD sequence was imported into the Galaxy platform. Sequence-processing operations were performed using SeqKit. The original sequence was retained as the reference sequence, while a second version was prepared containing the artificial mutation.

The sequences were translated so that the effect of the artificial nucleotide change could be examined at the protein level.

- Artificial Mutation

An artificial mutation was introduced into the DMD sequence. After translation, the modified protein contained a stop character within the protein sequence.

The relevant region of the original protein was:

AFYNQLQQLEQMT

The corresponding region of the modified protein was:

AFYNQLQ*LEQMT

The change therefore represents:

Q – stop codon (Q-*)

where Q represents glutamine and * represents a stop signal.

- Combining the Sequences

The original and modified protein FASTA datasets were concatenated in Galaxy. The resulting Dataset 32 contained two FASTA sequences in the following order:

1. Original DMD protein sequence
2. Modified DMD protein sequence

The two sequences were then examined directly to identify differences between the original and modified proteins.

- Multiple-Sequence Alignment Attempt

ClustalW was used in an attempt to perform a multiple-sequence alignment of the two protein sequences. The data type was set to PROTEIN, and FASTA was selected as the output alignment format.

However, the ClustalW job returned an error. Therefore, the final interpretation was based on direct comparison of the original and modified FASTA protein sequences using Concatenate on SeqKit results rather than on a successful ClustalW alignment.

Results

The concatenated FASTA dataset successfully contained two DMD protein sequences. The first sequence represented the original protein, while the second represented the artificially modified protein.

Most of the amino-acid sequences were identical. A difference was identified in the region:

Original:

`...NSRWIEFCQLLSERLWLEYQNNIAFYNNQLQQLEQMTTTAENW...`

Modified:

`...NSRWIEFCQLLSERLWLEYQNNIAFYNNQLQ\*LEQMTTTAENW...`

The comparison showed that the glutamine residue (Q) in the original sequence was replaced by a stop character (*).

This stop character occurs within the protein sequence and is therefore different from the normal terminal stop at the end of the translated sequence. The internal stop indicates premature termination of translation in the artificially modified protein.

Summary of the Sequence Difference

Feature	Original/WT	Artificially Modified
Relevant Sequence	AFYNNQLQQLEQMT	AFYNNQLQ*LEQMT
Change	Q retained	Q replace by *
Amino acid affected	Glutamine (Q)	Stop signal (*)
Translation	Continues through the region	Predicted to terminate prematurely
Expected protein	Full reference sequence	Truncated protein

WT vs. Mutant Protein Comparison

The original and modified sequences were nearly identical except for the artificially introduced stop signal. In the original protein, translation continues through the region containing the glutamine residue. In contrast, the modified protein contains an internal stop character.

Because translation is expected to terminate when a stop codon is encountered, the modified sequence is predicted to produce a shorter dystrophin protein. The remaining amino-acid sequence after the artificial stop would therefore not be translated into the final protein product.

The artificial mutation demonstrates how a single sequence alteration can produce a major change in the predicted protein product. Although the change itself involves only one amino-acid position in the translated sequence, its location as a premature stop can affect the length and potentially the function of the resulting protein.

Interpretation of the Artificial Mutation

The artificial Q – * change represents the protein-level consequence of introducing a premature termination signal. Unlike a substitution that simply changes one amino acid to another, a premature stop can terminate translation and prevent production of the complete protein.

For dystrophin, this type of sequence alteration is biologically important because functional dystrophin is required for the structural stability of muscle fibers. Pathogenic DMD variants that prevent production of functional dystrophin are associated with Duchenne muscular dystrophy.

The result of this experiment therefore supports the hypothesis. The artificially modified sequence is predicted to produce a truncated dystrophin protein compared with the original sequence.

However, the artificial mutation should not be interpreted as proof that this exact sequence change causes DMD in humans. The experiment was an in-silico demonstration and did not include patient data, protein-expression experiments, structural analysis, or functional testing.

Limitations

Several limitations should be considered in this study.

First, the mutation introduced in the project was artificial. It was created for the purpose of examining the predicted effect of a premature stop signal and was not presented as a patient-derived DMD mutation.

Second, the analysis was performed entirely using sequence-based bioinformatics methods. The predicted truncation of dystrophin does not directly demonstrate how the modified protein would behave inside a muscle cell.

Third, a ClustalW multiple-sequence alignment was attempted using the concatenated FASTA dataset, but the ClustalW job returned an error. Therefore, a successful alignment could not be included as evidence. Instead, the original and modified FASTA sequences were compared directly, which clearly revealed the Q-to-stop difference.

Finally, no protein-structure prediction, molecular simulation, gene-expression analysis, or laboratory experiment was performed. Therefore, the actual structural and functional effects of the artificial mutation cannot be determined from this analysis alone.

Conclusion

This study used a bioinformatics workflow to investigate the predicted effect of an artificial mutation in the DMD gene product. The DMD reference transcript NM_004006.3 was obtained

from NCBI and processed in Galaxy to produce an original protein sequence and an artificially modified protein sequence.

Direct comparison of the two protein sequences identified the change:

Original: `AFYNQLQQLEQMT`

Modified: `AFYNQLQ\*LEQMT`

This represents a Q-to-stop (Q - *) change that introduces a premature termination signal into the protein sequence. The modified sequence is therefore predicted to produce a truncated dystrophin protein.

The result supports the hypothesis that introducing a premature stop signal into the DMD coding sequence can substantially alter the predicted protein product. Although the attempted ClustalW alignment was unsuccessful, the artificial mutation was clearly identified through direct FASTA sequence comparison. The analysis demonstrates how bioinformatics tools can be used to examine the relationship between DNA sequence changes and predicted protein consequences.

References

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